Outlier Detection Method based on Improved K-means Clustering Algorithm

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ABSTRACT

Outlier detection is one of the important methods of data mining and analysis. The previous K-means outlier detection methods are easy to be misled by outliers in the clustering process and cannot detect outliers well. To solve this problem, we propose a method of outlier detection based on an improved K-means clustering algorithm. This method adaptively detects edge points (including outliers and edge points of other clusters) through the local threshold obtained from the distance distribution of the data points in the cluster to the centroid. The edge points are ignored when the centroid is updated to avoid the situation that the edge points mislead the centroid update. After completing the clustering, the farthest point in each cluster is put into the outlier cluster iteratively. When the average distance of the data in the cluster reaches stability or reaches the maximum number of iterations, the iteration ends. This method retains the ease of use of traditional K-means which accelerates the speed of clustering iteration and finds outliers accurately in data sets. Experimental results show that the algorithm can effectively detect outliers and have good results on multiple data sets.

CCS CONCEPTS

• Computing methodologies • Machine learning • Machine learning approaches

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KEYWORDS

Machine learning, Clustering algorithm, K-means, Outlier detection

1 Introduction

The purpose of outlier detection is to detect unusual objects or objects that are inconsistent with most objects in some respects. These outlier objects are called outliers. The methods of outlier detection come from the fields of statistical learning, machine learning, and data mining, which have important application value in medical, fraud detection, industrial anomaly detection, intrusion detection, and other fields. In reality, the data sets with labels are rare, so unsupervised learning is more widely used. Outlier detection can be regarded as an extreme clustering problem. The traditional clustering algorithm does not take into account the outliers in the data set. In the clustering process, outliers may affect the clustering results. Therefore, the traditional clustering algorithm is inappropriate for outlier detection.

1.1 Related Work

Clustering is one of the basic methods of data mining. The traditional K-means clustering algorithm [1] is a widely used clustering technique. There are still many related algorithms that have been studied but with shortcomings [2][3][4][5]. The choice of the initial centroid has a great influence on it and it requires the number of clusters to be provided. The K-means algorithm does not consider the existence of outliers in the data sets during the clustering process and the outliers will affect its clustering effect. Therefore, the K-means algorithm needs to be modified for outlier detection.

The existing techniques of applying the K-means clustering algorithm to deal with outliers mainly include two types. One is to

prevent outliers from being selected as the initial centroid. The other is to add a data processing method based on the traditional K-means clustering algorithm. The most widely used technique is K-means++ [6] which is to make the distance between the centroids of the initial clusters as far as possible. In 2012, Bahmani et al. proposed a scalable alternative Scalable K-means++ [7]. The algorithm obtains an initial center set that can be proved to be close to the optimal solution, but the influence of outliers on the objective function is inevitable. In 2017, Peter et al. proposed the method K-mean sharp（K-means#） for outlier detection during the clustering process [8]. Because classic algorithms of K-means clustering are easily misled by outliers, the paper modified its centroid update step to avoid the influence of outliers when calculating the new centroid. Automatic detection of outliers through a global threshold derived from the distance distribution of the point to the centroid. It requires neither user intervention nor prior knowledge of the number of outliers. Since K-means# retains the structure of K-means, it inherits the ease of use of k-means. The algorithm performs outlier detection and clustering at the same time. It is possible to put edge points in other clusters into outlier clusters, which will affect the effect of outlier detection.

In 1997, Brito et al. proposed a test method to detect clustering structure and outliers [9]. In 2000, Breunig et al. proposed a density-based outlier detection algorithm [10]. The algorithm does not explicitly divide data points into normal points or outliers but calculates local outlier factors (LOF) for each data point to measure the outlier degree of each data point. The local outlier factor of a data point is the ratio of the local density to the local density of its neighboring area. The algorithm relies on the ratio of outliers and the number of neighbors provided by the user. In 2003, He et al. proposed the FindCBLOF algorithm [11]. The algorithm firstly divides the data sets into many large clusters and small clusters. Then it uses the FindCBLOF algorithm to calculate the local outlier factor for each data point and finally uses the local outlier factor to measure the data point degree of the outlier. In 2019, Zhao et al. propose a framework — called Locally Selective Combination in Parallel Outlier Ensembles (LSCP) — which addresses the issue by defining a local region around a test instance using the consensus of its nearest neighbors in randomly selected feature subspaces [12]. The top-performing basing on detectors in this local region are selected and combined as the model's final output. In 2020, Li et al. present a novel outlier detection algorithm called COPOD [13]. COPOD firstly constructs an empirical copula and uses it to predict tail probabilities of each given data point to determine its level of extremeness. COPOD is both parameter-free, highly interpretable, and computationally efficient.

1.2 Our Contribution

Through the analysis of the existing cluster-based outlier detection algorithm, it is known that performing outlier detection in the clustering process may regard the edge points of other clusters as outliers, resulting in the inaccuracy of the detected outliers. The initial centroid of the K-means clustering algorithm is randomly selected. When the initial centroid is too biased, the edge points will affect the clustering results. Normal points at the edge of other clusters and outliers in the data set. In order to solve the problem that the edge points have an impact on the traditional K-means clustering algorithm. This paper proposes a method of ignoring the edge points in the cluster and detecting outliers in the cluster during the clustering process. This paper proposes a method of ignoring the edge points in the cluster during the clustering process and detecting outliers in the cluster at the end of the clustering. Removing the edge points during the clustering process helps to find the centroid quickly, and detecting outliers on the clustered clusters will improve the accuracy. According to the characteristic that the data in the normal cluster is more concentrated and the data in the abnormal cluster are more sparse, the algorithm ends when the average distance of the data points in the cluster stabilizes. The method in this paper retains the basic structure of traditional K-means and its ease of implementation and high efficiency. By modifying the steps of centroid update, the speed of clustering iteration is accelerated and the accuracy of outlier detection is improved.

2 Outlier Detection Method based on Improved K-means Clustering

2.1 Related Theories

As one of the classic clustering algorithms, K-means is widely used because of its simplicity and ease of implementation [1]. The main idea of the algorithm is to cluster the data into k clusters to make the data points in each cluster more similar. Given a dataset $X = \{x_1, x_2, \dots, x_n\}$, the purpose is to cluster n data points into k $(k \le n)$ clusters $S_1, S_2, \dots, S_k$ to minimize the variance within the cluster. The goal is to find:

$$\underset{S}{argmin} \sum_{i=1}^{k} \sum_{x \in S_i} \|x - \mu_i\|_2^{\,2} \tag{1}$$

where μ_i is the mean value of the data points in set S_i,

$$\mu_i = \frac{1}{|S_i|} \sum_{x \in S_i} x \tag{2}$$

The cluster is defined as follows:

$$S_i = \{x: \|x - \mu_i\|_2^{\,2} \le \|x - \mu_j\|_2^{\,2} \forall j, 1 \le j \le k\} \tag{3}$$

$S_i \cap S_j = \emptyset, \forall 1 \le i, j \le k$. Each data point can only be assigned to one cluster.

The basic steps of the algorithm are as follows:

1) Initialization: Randomly select a set of initial centroids $\mu_i^{(0)} (1 \le i \le k)$ in the data set.

2) Data point allocation: Use formula 3 to cluster into k clusters.

3) Centroid update: Use formula 2 to calculate the mean value of the data points in each cluster as the new centroid of the cluster.

4) When the centroid does not change or reaches the maximum number of iterations, stop the iteration; otherwise, go to step 2.

Although the traditional K-means clustering algorithm is simple and effective for data clustering, it will affect the result of data clustering when there are outliers in the data set. This article modified the steps of K-means algorithm centroid update. The local threshold obtained by the distance distribution from the data

points in the cluster to the centroid is used to detect edge points adaptively, and the edge points are ignored when the centroid is updated. The purpose is to avoid the situation where edge points affect the clustering results.

2.2 Outlier Detection Method based on Improved K-means Clustering

The algorithm uses the centroid to define the prototype where the centroid is the mean value of a set of data points. Since outliers in the data set and edge data points of other clusters will affect the clustering results and the speed of centroid iteration during the clustering process, we propose an outlier detection method that improves the K-means clustering algorithm. Based on the structure of the traditional K-means clustering algorithm, this method changes the centroid update step and adds the function of finding outliers in the data set. In this section, we will show how the improved K-means clustering algorithm loses edge points during each iteration, avoiding their influence on the centroid update, and how to find outliers after clustering.

2.2.1 Centroid update

Set S is a cluster, the standard deviation of S is defined as follow:

$$\sigma = \sqrt{\frac{1}{|S|}\sum_{x\in S}\left(x - \frac{1}{|S|}\sum_{x\in S} x\right)^2} \tag{4}$$

The edge points of S is defined as follow:

$$S' = \{x: x \geq 3\sigma, x \in S\} \tag{5}$$

Firstly, calculate the standard deviation in each cluster (As shown in formula 4), then remove the edge points to get cluster $\tilde{S} = S¥S'$. Compared with the centroid update step in the traditional K-means clustering algorithm, the purpose of this method is to quickly determine the centroid and avoid the influence of edge points when the centroid is updated.

Algorithm 1: Improved K-means Clustering Algorithm

Input: dataset $X = \{x_1, x_2, \dots, x_n\}$, number of clusters k, the maximum number of iterations t.

Output: cluster $S_1, S_2, \dots, S_k$.

Step 1. Randomly select k initial centroids in the data set.
Step 2. Use formula 3 to cluster into k clusters.
Step 3. Use formula 4 to calculate the standard deviation of each cluster. Use formula 5 to define data points in the cluster that exceed 3σ as edge points. After removing the edge points in the cluster, use formula 2 to calculate the mean value of the data points in the cluster to obtain a new centroid.
Step 4. When the centroid no longer changes or the number of iterations reaches the maximum, go to step 5; otherwise, go to step 2.
Step 5. Return $S_1, S_2, \dots, S_k$ (this cluster contains edge points).

2.2.2 Outlier detection

Put the data point farthest from the centroid in cluster S_i into outlier cluster S_i'', S_i'' is defined as follows:

$$S_i'' = \{x | d_{max}(x, \mu) = \max\{\|x - \mu\|_2, x \in S_i\}\} \tag{6}$$

Cluster $\hat{S}_i$ refers to removing the farthest point in each cluster S_i, which is defined as follows:

$$\hat{S}_i = S_i \backslash S''_i \tag{7}$$

Merge the outliers in each cluster S_i'' to the outlier cluster S'', S'' is defined as follows:

$$S'' = \cup_{i=1}^{k} S_i'' \tag{8}$$

The outlier clusters S'' of the previous m iterations are merged into outlier clusters S_{merge}, S_{merge} is defined as follows:

$$S_{merge} = \cup_{m=1}^{t} S''^{(m)} \tag{9}$$

$S''^{(m)}$ is the set of outliers generated in the *m-th* iteration.

After removing the outliers, the normal clusters will become more and more compact, while the outliers will be scattered. When the average distance of the cluster becomes stable, the outliers in the normal cluster are basically removed. The average distance of the cluster is defined as follows:

$$C = \frac{1}{k+1}\left(\sum_{i=1}^{k}\frac{1}{|\hat{S}_i|(|\hat{S}_i|-1)}\sum_{x_j, x_l \in \hat{S}_i}\|x_j - x_l\|_2 + \frac{1}{|S_{merge}|(|S_{merge}|-1)}\sum_{x_j, x_l \in S_{merge}}\|x_j - x_l\|_2\right) \tag{10}$$

Where $x_j \neq x_l$.

Algorithm 2: Outlier Detection Algorithm based on Improved K-means

Input: cluster $S_1, S_2, \dots, S_k$, threshold ε, the maximum number of iterations n.

Output: outlier cluster S_{merge}.

Step 1. Use formula 2 to calculate the centroid of each cluster.
Step 2. Use formula 6 to put the data point farthest from the center of mass in the cluster into the outlier cluster, and use formula 7 to remove it from the normal cluster. Use formulas 8 and 9 to merge the outliers generated in each iteration into the outlier cluster S_{merge}.
Step 3. Use formula 10 to calculate the average distance of the clusters.
Step 4. When the average distance of the cluster stabilizes or reaches the maximum number of iterations, go to step 5; otherwise, go to step 2.
Step 5. Return outlier cluster S_{merge}.

3 Experiment

3.1 Data Set

This experiment uses public data sets and artificial data sets. The public data set comes from the ODDS website, and the artificial data set is created using the data set the tool in sklearn. This paper proposes an outlier detection method based on an improved K-means clustering algorithm. This method inherits the characteristics of the K-means algorithm and has better performance on low-dimensional convex data sets. Therefore, we created four sets of two-dimensional data sets (d1, d2, d3, d4) to verify the performance of this method. The four sets of data are composed of 1800 normal points and 200 outliers. The variance of

the normal point part of the d1 data set is a Gaussian distribution of 0.5 and 0.5, the variance of the normal point part of the d2 data set is a Gaussian distribution of 0.5, 0.5 and 0.5, and the variance of the normal point part of the d3 data set is 0.5, 0.5, 0.5 and 0.5 Gaussian distribution, the variance of the normal point part of the d4 data set is a Gaussian distribution of 0.3 and 1.5. ODDS data sets are mostly high-dimensional data. In order not to affect the experimental effect, we must reduce the dimensionality of the data before using the method in this paper. The dimensionality reduction method we use is principal component analysis [14]. The K-means clustering algorithm needs to set the value of k (the number of clusters) in advance. The elbow rule [14] can be used when the k value of the data set cannot be determined. The specific information of the public data set is shown in Table 1.

Table 1 Public data set information table

Data set	Number of data	Data dimension	Data dimension (Dimensionality reduction)	Number of outliers	k value
d5 (mammography)	11183	6	5	260	2
d6 (thyroid)	3772	6	3	93	3
d7 (satellite)	6435	36	4	2036	3
d8 (satimage-2)	5803	36	4	71	3
d9 (wbc)	278	30	8	21	3
d10 (glass)	214	9	4	9	2

3.2 Evaluation Metrics

The evaluation metrics used in the experiment of this paper are the AUC value (the area enclosed by the ROC curve), accuracy, precision, recall, and F1-score. For a given data set $X = \{x_1, x_2, \dots, x_n\}$ and real category $Y = \{y_1, y_2, \dots, y_n\}$, the samples are divided into four situations according to the combination of the real category and the outlier detection category, as shown in Table 2.

Table 2 Confusion matrix

Real category	Outlier detection category	
	Positive	Negative
Positive	TP	FN
Negative	FP	TN

True Positive (TP) refers to the point that is detected as an outlier and its true category is also an outlier. False Negative (FN) refers to points that are detected as normal but its true category is an outlier. False Positive (FP) refers to points that are detected as outliers but its true category is normal. True Negative (TN) refers to the point that is detected as normal and its true category is also normal. Outlier detection can be regarded as an extreme data imbalance problem, so this paper uses the commonly used metrics for classification learning to evaluate the algorithm. Accuracy is defined as the proportion of the number of samples correctly clustered in each category to the total number of samples. Precision means the proportion of correctly clustered samples among all the predicted positive samples. The recall rate means the proportion of correctly clustered samples among all positive samples. To comprehensively consider the precision and recall rate, this paper adopts the F1-score. The precision (P), recall (R), and F1-score (F1) are defined as:

$$P = \frac{TP}{TP+FP} \tag{11}$$

$$R = \frac{TP}{TP+FN} \tag{12}$$

$$F1 = \frac{2 \times P \times R}{P+R} \tag{13}$$

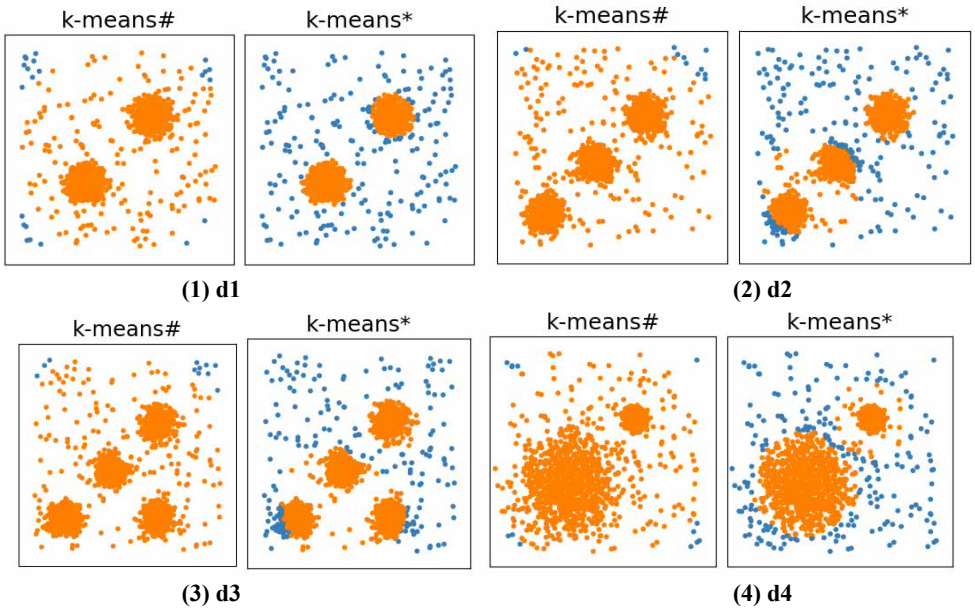


Figure 1 Experimental comparison of artificial data sets

It is not comprehensive to evaluate the performance of the method in this paper simply using static metrics such as accuracy, precision, and recall. For outlier data sets, it is more appropriate to use the ROC measurement standards of FPR and TPR on the horizontal and vertical axes. This paper uses the AUC value of the area under the ROC curve. FPR and TPR are defined as:

$$FPR = \frac{FP}{FP+TN} \tag{14}$$

$$TPR = \frac{TP}{TP+FN} \tag{15}$$

3.3 Experiment and Result Analysis

First of all, the experimental method (denoted as K-means*) in this paper will be compared with the K-means# algorithm. Four sets of two-dimensional artificial data sets are used to verify the effectiveness of the outlier detection method in this paper. The experimental results are shown in Figure 1. The orange points are normal values and the blue points are outliers. It can be seen from the figure that the method in this paper can find outliers in the data set well, achieve the expected effect, and prove that the algorithm is true and effective.

The algorithm parameter settings are shown in Table 3.

Table 3 The parameter settings of algorithm

Algorithm	Parameter settings
LOF	$k = 35$
K-means#	The maximum number of iterations is 50
COPOD	-
K-means*	The maximum number of iterations is 100, $\varepsilon = 0.00001$

For a total of 10 data sets of artificial data sets and public data sets, compare the method in this paper (abbreviated as KM* in the table) with the three outlier detection algorithms LOF, COPOD, and K-means# (abbreviated as KM# in the table). The implementation of LOF and COPOD can be obtained in the PyOD library. The experimental results are shown in Table 4. On the d1, d2, and d4 data sets, the overall performance of the method in this paper performs well, and only the accuracy rate is inferior to K-means#; on the public data set, it can be seen from the average value of each evaluation metrics. Compared with the other three algorithms, the performance of the method in this paper is slightly improved and the result is stable. The above experiments also show that the algorithm inherits the characteristics of K-means, is simple and easy to use, and performs well on low-dimensional convex data sets.

Table 4 Comparison of experimental results of five algorithms

		d1	d2	d3	d4	***AVG***	d5	d6	d7	d8	d9	d10	***AVG***
AUC	LOF	0.847	0.817	**0.839**	0.769	0.818	0.732	0.913	0.562	0.438	0.940	**0.771**	0.726
	KM#	0.555	0.528	0.528	0.533	0.537	0.750	0.548	0.637	0.903	0.870	0.562	0.717
	COPOD	0.727	0.720	0.701	0.634	0.700	**0.806**	0.841	0.612	0.916	0.836	0.464	0.746
	KM*	**0.954**	**0.854**	0.824	**0.816**	**0.862**	0.782	**0.931**	**0.671**	**0.917**	**0.946**	0.752	**0.833**
accuracy	LOF	0.945	0.934	**0.942**	0.917	0.934	**0.964**	0.962	0.612	**0.977**	**0.942**	0.925	0.897
	KM#	0.911	0.906	0.906	0.907	0.907	0.915	0.946	0.769	0.945	0.880	0.873	888
	COPOD	0.864	0.889	0.894	0.869	0.879	0.913	0.913	0.752	0.915	0.907	0.907	0.885
	KM*	**0.982**	**0.946**	0.935	**0.932**	**0.949**	0.958	**0.971**	**0.785**	0.964	0.916	**0.938**	**0.922**
precision	LOF	0.847	0.817	0.839	0.769	0.818	**0.598**	**0.608**	0.552	0.522	**0.731**	0.536	0.591
	KM#	**0.955**	**0.952**	**0.952**	**0.953**	**0.953**	0.570	0.534	**0.859**	**0.582**	0.645	0.527	0.620
	COPOD	0.668	0.704	0.712	0.641	0.618	0.576	0.576	0.810	0.572	0.714	0.488	0.623
	KM*	0.947	0.848	0.818	0.810	0.856	0.583	0.614	0.837	0.579	0.714	**0.539**	**0.644**
recall	LOF	0.847	0.817	**0.839**	0.769	0.818	0.598	0.608	0.552	0.522	0.731	0.536	0.591
	KM#	0.555	0.528	0.528	0.533	0.536	0.750	0.548	0.637	0.903	**0.870**	0.562	0.712
	COPOD	0.727	0.720	0.701	0.634	0.700	**0.806**	**0.841**	0.612	0.916	0.836	0.464	**0.746**
	KM*	**0.954**	**0.854**	0.824	**0.816**	**0.862**	0.738	0.598	**0.646**	**0.918**	0.867	**0.573**	0.723
F1_score	LOF	0.847	0.817	**0.839**	0.769	0.818	0.598	0.608	0.552	0.522	0.731	0.536	0.591
	KM#	0.576	0.527	0.527	0.536	0.542	0.597	0.539	**0.644**	0.624	0.689	0.530	0.604
	COPOD	0.690	0.712	0.706	0.637	0.686	0.606	0.606	0.610	0.590	**0.756**	0.476	0.607
	KM*	**0.950**	**0.850**	0.821	**0.813**	**0.859**	**0.615**	**0.622**	0.639	**0.633**	0.719	**0.543**	**0.629**

The following experiment compares the effects of different values of ε and the maximum number of iterations on the detection results of artificial data sets. First, we fixed the maximum number of iterations to 50, and the experimental results are shown in Table 5. Secondly, we fixed ε to 10^{-4}, and the experimental results are shown in Figure 6. After experimental verification, we chose $\varepsilon = 10^{-4}$ and the maximum number of iterations is 100.

Table 5 The effect of ε on the results

	AUC			accuracy			precision			recall			F1_score		
	10^{-3}	10^{-4}	10^{-5}	10^{-3}	10^{-4}	10^{-5}	10^{-3}	10^{-4}	10^{-5}	10^{-3}	10^{-4}	10^{-5}	10^{-3}	10^{-4}	10^{-5}
d1	0.760	0.760	0.760	0.952	0.952	0.952	0.975	0.975	0.975	0.760	0.760	0.760	0.829	0.829	0.829
d2	0.743	0.743	0.743	0.946	0.946	0.946	0.944	0.944	0.944	0.743	**0.808**	0.743	0.808	**0.839**	0.808
d3	0.727	**0.743**	0.727	0.940	**0.946**	0.940	0.914	**0.944**	0.914	0.727	**0.743**	0.727	0.786	**0.808**	0.786
d4	0.713	0.713	0.713	0.935	0.935	0.935	0.888	0.888	0.888	0.713	0.713	0.713	0.769	0.769	0.769
AVG	0.736	**0.740**	0.736	0.943	**0.945**	0.943	0.930	**0.938**	0.930	0.736	**0.756**	0.736	0.798	**0.811**	0.798

Table 6 The influence of the maximum number of iterations on the results

	AUC			accuracy			precision			recall			F1_score		
	100	80	60	100	80	60	100	80	60	100	80	60	100	80	60
d1	**0.954**	0.904	0.810	**0.982**	0.980	0.962	0.974	**0.984**	0.980	**0.954**	0.904	0.810	**0.950**	0.940	0.872
d2	**0.854**	0.807	0.779	0.946	0.945	**0.951**	0.848	0.867	**0.932**	**0.854**	0.807	0.779	**0.851**	0.834	0.835
d3	**0.824**	0.791	0.740	0.935	**0.939**	0.937	0.818	0.847	**0.872**	**0.824**	0.791	0.741	**0.821**	0.816	0.788
d4	**0.816**	0.774	0.740	0.932	0.774	**0.937**	0.81	0.827	**0.872**	**0.816**	0.774	0.741	**0.813**	0.798	0.788
AVG	**0.862**	0.819	0.767	**0.949**	0.910	0.947	0.856	0.881	**0.914**	**0.862**	0.819	0.768	**0.859**	0.847	0.821

4 Summary

In order to avoid the outliers affecting the clustering effect during the traditional clustering process, and study the impact of the outlier detection, this paper proposes an outlier detection method based on an improved K-means clustering algorithm. This method adaptively detects edge points based on the local domain value obtained from the distance distribution of the data points in the cluster to the centroid. The edge points are ignored when the centroid is updated, in order to avoid the situation that the edge points mislead the centroid update and speed up the iterative process. In order to avoid misclassification in the clustering process and improve the performance of outlier detection, outlier detection is performed on the cluster after the clustering is completed. The experiments prove the effectiveness of the algorithm on artificial data sets and multiple public data sets. Comparative experiments on multiple public data sets show that the algorithm has good detection performance. It is close or better to the commonly used outlier detection algorithm in multiple application scenarios, which has certain adaptability and application value.

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